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NO_x FORMATION IN TURBULENT LEAN-PREMIXED COMBUSTION WITH MINIMUM HEAT LOSSES

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The formation of NO_x in turbulent combustion of lean-premixed methane/air mixtures and hydrogen/air mixtures was investigated at various levels of turbulent intensity using an impinging jet reactor. Much attention was paid to minimizing heat losses from the combustion field to the ambient. Experiments were carried out at atmospheric pressure and an inlet temperature of 323 K. The equivalence ratio ranged from 0.5 to 1.0 for CH_4/air mixtures and from 0.6 to 0.7 for H_2/air mixtures. Detailed results obtained at combustion gas temperatures (T_g) of 1390–2200 K with complete combustion are presented.

It was found that a change in the level of turbulence intensity did not result in a change in NO_x emissions. At $T_g > 1700$ K, the NO_x formation was contributed primarily by the thermal mechanism. The NO_x emissions were proportional to the reactor residence time and sensitive to the combustion gas temperature. The NO_x concentration for CH_4/air can be expressed as follows:

$$[NO_x] = \tau(5.96 \times 10^7) \exp(-31{,}500/T_g) \text{ (ppm)}$$

where τ is the mean residence time in ms and T_g in K. At $T_g < 1700$ K, however, the NO_x concentrations were independent of residence time and were only slightly sensitive to T_g. The NO_x concentrations for CH_4/air mixtures were in agreement with those for H_2/air mixtures. This implies that the NO_x was formed by a mechanism without hydrocarbon–nitrogen path. The nitrous oxide mechanism is expected to be an important mechanism for NO_x formation in the present impinging jet reactor.

Introduction

The emissions of oxides of nitrogen (NO_x) from combustion processes cause environmental pollution. An understanding of NO_x-formation mechanisms is essential for reducing the emissions levels. The chemistry of NO_x formation in a laminar laboratory flame has been studied for a long time, and the major reaction mechanisms are well understood [8]. In most practical combustors and engines, however, combustion proceeds in turbulent flow fields. The NO_x formation in turbulent combustion has been studied in many aspects [1–7]. An investigation of the NO_x formation in impinging jet combustion of premixed mixtures in a closed chamber demonstrated that a significant reduction of NO_x emissions (up to 88%) was attained at high turbulence levels [7]. However, the reduction cannot be fully attributed to the enhanced mixing of burned and unburned gases since heat losses to the walls of the chamber should be increased with the increased turbulence levels. In contrast with the considerable reduction of NO_x emissions, H. Semerjian et al. [2] reported that the rates of NO_x formation in fuel-lean, turbulent flames were higher when compared with those found for laminar flames at the same temperature. At present, the effect of turbulence on NO_x formation remains unclear.

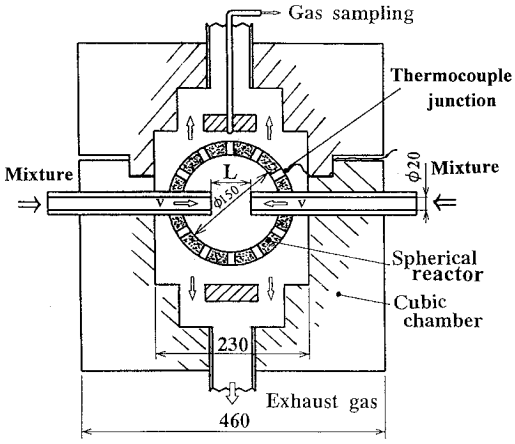

FIG. 1. Schematic of impinging jet reactor.

The application of the lean combustion concept to combustor design is an area of active investigation because of the potential benefit of ultra-low NO_x emissions [1]. In fuel-lean combustion, the contribution of the Zeldovich mechanism and the Fenimore mechanism to NO_x formation becomes small when the combustion temperature is below 1800 K [6]. It is proposed that most of the NO_x is formed by the nitrous oxide mechanism [5,6]. However, little experimental data are available on evaluation of the mechanisms.

In this study, the characteristics of NO_x formation in lean-premixed combustion were investigated in an impinging jet reactor made of ceramics. Much attention was paid to minimizing heat losses from the combustion field to the ambient. Detailed results obtained at combustion gas temperatures of 1390–2200 K with complete combustion are presented in this paper. The effects of turbulent intensity on NO_x emissions are examined.

Experimental Apparatus

The impinging jet reactor used in this study is shown in Fig. 1. The spherical ceramic reactor, 150 mm in inner diameter, is fed by two ceramic tubes of 20 mm in diameter. The reactor has 40 holes with diameters of 12 mm equally spaced on its wall. Inside the reactor, the two jets of premixed mixture impinge at the same velocity and form a turbulent flow field where combustion takes place. The burned gas goes through the holes and the space between the reactor and the heat-insulating firebrick chamber, of 230-mm cube, and finally is exhausted. The heat losses from the reactor are reduced greatly since the reactor is surrounded by hot, burned gases.

Experiments were conducted at atmospheric pressure for CH_4/air and H_2/air mixtures heated electrically to a temperature of 323 K. The equivalence ratio (ϕ) ranged from 0.5 to 1.0 for CH_4/air mixtures and from 0.6 to 0.7 for H_2/air mixtures. For H_2/air mixtures, flashback into the tubes occurred at combustion gas temperatures higher than 1700 K. Thus, the range of combustion gas temperatures was narrowly limited in the present measurement. The reactor mean residence time (τ) changed from 5.3 to 35.0 ms. H_2/air mixtures were used to examine the contribution of hydrocarbon–nitrogen paths to NO_x formation at lower combustion gas temperatures. The characteristics of NO_x formation for H_2/air mixtures were compared with those for CH_4/air mixtures.

To generate various levels of turbulence in the reactor, the jet velocity (V) and the distance between the exit planes of the two tubes (L) were varied. Data were collected for L = 20, 40, and 150 mm, and V = 20, 40, and 80 m/s, corresponding to mass flow rates of mixture (Q_m) of 13.8, 27.6, and 55.2 g/s, respectively.

Combustion gas temperatures were measured with Pt-Rh 20%/Pt-Rh 40% thermocouples coated with silicate-based zirconia at the center of several holes on the reactor wall. The measured temperatures were not found to differ significantly from each other. The averaged values of the measured temperatures were referred to as combustion gas temperature (T_g). The temperatures of the inner and outer surfaces of the reactor and those of cubic chamber walls were also measured. On the basis of these surface temperatures, the heat loss from the reactor was estimated to be less than approximately 8% of the heat release in the reactor. The combustion gas temperatures measured were only 50–110 K lower than the corresponding adiabatic temperatures. The turbulence characteristics of the impinging jet flow were measured under nonburning conditions by using a hot-wire anemometer. A 5-μm wire probe, calibrated in an orifice wind tunnel, was used.

Burned gas was withdrawn using a water-cooled stainless steel gas sampling probe with a 1-mm-diameter orifice and then fed to a gas analyzer to determine the concentrations of NO, NO_x, CO, CO_2, O_2, and total hydrocarbons. In the analyzer, NO and NO_x were detected by a chemiluminescence analyzer, CO and CO_2 by nondispersive infrared detectors, the total hydrocarbons by a flame ionization detector, and O_2 by a paramagnetic detector. The sampling line that connected the probe to the gas analyzer was heated electrically to prevent condensation of water vapor. The temperature and concentration data presented in this paper were collected after the establishment of thermally steady states at each reactor condition. The values of NO_x emissions, obtained in this study and derived from published papers, were expressed in terms of ppm corrected to 0% O_2, wet basis.

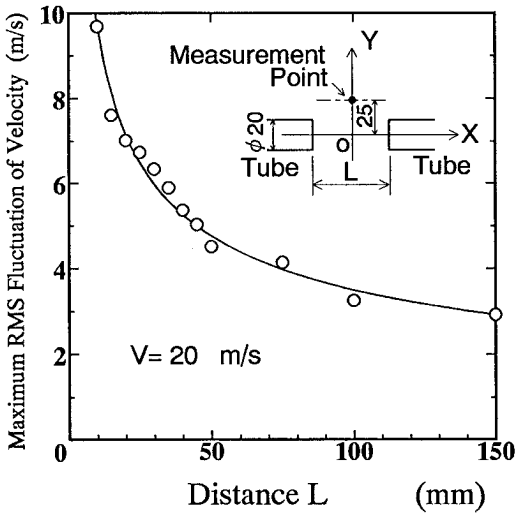

FIG. 2. Variation of maximum rms fluctuation of velocity with separation distance L.

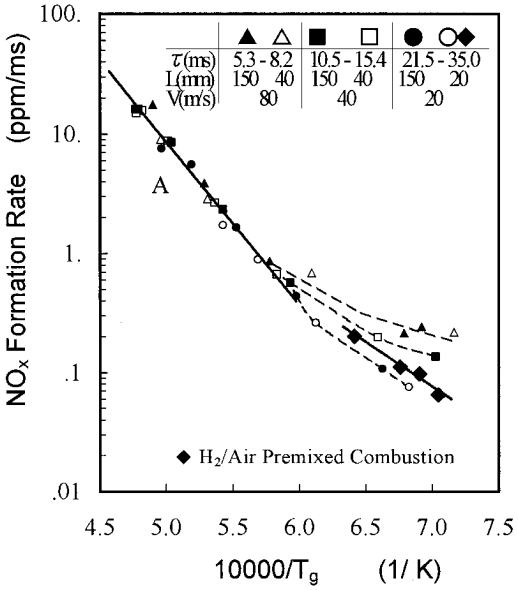

FIG. 3. Dependence of NO_x formation rate for CH_4/air and H_2/air on combustion gas temperature.

Experimental Results

In this work, the impingement strength of the two counterflow jets in the reactor was changed by varying L and V. The results of turbulence measurements revealed that an increase in V and a decrease in L intensified the turbulence in the bulk of the impinging flow field. In addition, it was found that the root-mean-square fluctuations of gas velocity (u') gradually increased from the tube exit toward the stagnation plane perpendicular to the axis of the tubes. On the stagnation plane, u' reached a peak value at a radial position of 1.25 times the tube diameter. Figure 2 illustrates the variation of u' at the peak with respect to L. A decrease in L from 150 to 10 mm resulted in an increase in the peak value from 2.9 to 9.7 m/s when V was equal to 20 m/s.

Figure 3 shows the dependence of NO_x formation rates on combustion gas temperatures for various turbulence levels. The NO_x formation rate on the vertical axis was defined as the ratio of NO_x concentration to τ. The values of τ can be divided into three groups in terms of V. Each test condition specified by V and L is labeled with a symbol in the figure corresponding to a level of turbulence intensity. The data corresponding to those in the figure are listed in Table 1.

Figure 3 shows that when $T_g > 1700$ K ($10^4/T_g < 5.88$), the NO_x-formation rate for CH_4/air mixtures depended primarily on T_g and was independent of V and L. Under the operating conditions in this study, therefore, the turbulence intensity generated by the impinging jets did not change the rate of NO_x formation.

The linear line (A) fitted to the data of NO_x emissions at temperatures higher than 1700 K was represented by the following equation:

$$[NO_x] = \tau\,(5.96 \times 10^7)\exp(-31{,}500/T_g)\ \text{(ppm)}$$

where τ is in ms and T_g in K. The overall activation energy for NO_x formation was estimated to be 63 kcal/mol by the expression based on the Arrhenius relationship.

Line A was compared with the data for NO_x-formation rates derived from published papers. H. Semerjian et al. investigated NO_x formation in lean-premixed, turbulent flames of prevaporized Jet-A fuel stabilized by perforated plate flame holders [2]. The NO_x was found to be formed more quickly in the recirculation zone behind the flame holders than in the postflame zone. The NO_x-formation rates, derived with respect to the residence time in the combustion reaction zone, are shown in Fig. 4. The data on line A were found to be very close to those of H. Semerjian et al., as well as those obtained in a jet-stirred combustor burning CH_4/air mixtures by C. J. Jachimowski et al. [4]. Figure 4 also shows another set of data (line B) for the NO_x-formation rates obtained by R. C. Steele et al. in a fuel-lean, jet-stirred reactor [6]. Although the NO_x-formation rates were greater by a factor of 3 to 4 when compared with line A, the overall activation energy of 55 kcal/mol, obtained by fitting to the data, was close to that for line A.

The data for the fuel-lean combustion with intense back mixing, shown in Fig. 4, are in agreement on the NO_x-formation rates being proportional to the

TABLE 1
Emissions data for combustion of CH_4/air and H_2/air mixtures

	T_g (K)	τ (ms)	NO_x (ppm)	CO (ppm)	$EI(NO_x)$ (g/kg)		T_g (K)	τ (ms)	NO_x (ppm)	CO (ppm)	$EI(NO_x)$* (g/kg)
	1395.6	8.22	1.8	82	0.09		1422.9	15.36	2.1	71	0.11
	1445.2	7.82	1.9	72	0.10		1517.3	14.57	2.9	73	0.16
	1473.0	7.44	1.6	76	0.08		1686.9	12.98	7.4	120	0.34
	1731.7	6.35	5.5	82	0.25		1715.8	12.93	8.6	102	0.40
CH_4	1883.9	6.10	17.6	104	0.71	CH_4	1844.3	11.92	28.0	180	1.09
V = 80 m/s	1891.7	5.82	22.8	170	0.88	V = 40 m/s	1863.6	11.90	31.8	156	1.30
	2016.1	5.69	51.5	584	1.85		1985.2	11.05	94.0	380	3.26
	2041.0	5.39	94.6	840	3.24		1994.7	11.14	98.0	740	3.61
	1465.2	35.01	2.7	4	0.15		2074.5	10.69	168.0	2340	5.59
	1509.0	28.84	3.1	30	0.15		2087.8	10.52	171.0	2580	5.41
	1633.0	31.75	8.3	23	0.40		2093.2	10.59	158.0	5180	5.19
	1673.5	26.03	11.4	71	0.49		2096.2	10.45	168.0	4620	5.20
CH_4	1758.4	29.57	26.2	37	1.11		1418.2	33.23	2.2		
V = 20 m/s	1809.1	24.10	39.9	120	1.46	H_2	1446.0	29.17	2.9		
	1842.7	28.87	49.8	113	1.93	V = 20 m/s	1477.8	26.35	2.9		
	1926.7	22.72	126.0	322	4.17		1556.2	26.48	5.4		
	2013.1	21.47	161.6	3050	4.98						

*$EI(NO_x)$: NO_x emission index

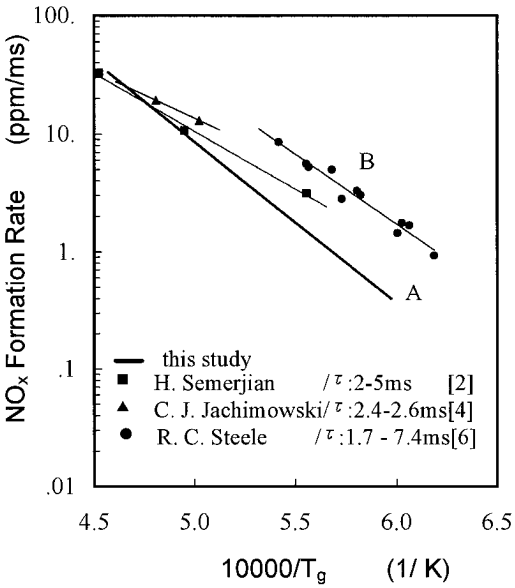

FIG. 4. NO_x formation rate for CH_4/air compared with data from previous experimental studies.

residence time and the values of the overall activation energy of NO_x formation approximately cover the range of 40–65 kcal/mol. These values were much lower than 134 kcal/mol for the thermal NO_x formation in a pure oxygen/nitrogen system but close to the heat of formation of atomic oxygen (59 kcal/mol) [10].

The mechanism of NO_x formation in laminar premixed combustion has been well understood [8] as a result of numerous experimental and modeling studies. Generally, NO_x formation by the thermal mechanism is proportional to its production time and depends heavily on the combustion gas temperature. The prompt NO_x formation occurs because of the superequilibrium concentrations of radicals (such O and OH) in regions near flame zone [8] or because of the Fenimore mechanism, especially at fuel-rich conditions [9]. In the present fuel-lean, turbulent, impinging jet combustion, the NO_x formation at the higher temperatures was primarily by the thermal mechanism; the Fenimore mechanism was not an important source of NO_x. In the turbulent flames, the intense back mixing between the burned and unburned gases occurred, which may lead to the rapid mixing of the excessive radicals from burning gases with unburned gases and then accelerate NO_x formation by the thermal mechanism at higher temperatures [8]. The difference between the overall activation energies for the turbulent flames and those for the pure oxygen/nitrogen system is prob-

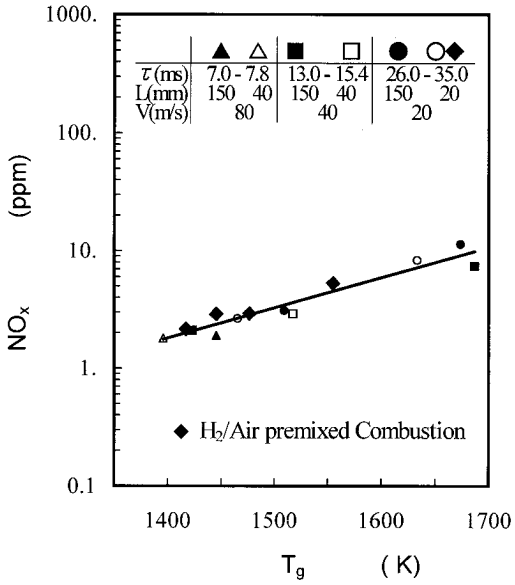

FIG. 5. NO_x emissions for CH_4/air and H_2/air at low-combustion gas temperatures.

ably accounted for by the rapid radical mixing in the flames. However, it is difficult to recognize essentially how the mixing affected the NO_x formation on the basis of the present experimental data.

The data for the NO_x-formation rates at temperatures below 1700 K deviated from line A, as shown in Fig. 3. The formation rates depended on the residence time in the reactor: The shorter the residence time, the greater the formation rate at a given T_g. The measured NO_x concentrations, rather than the NO_x formation rates, were plotted versus combustion gas temperature in Fig. 5. It is clear that the NO_x concentrations were independent of the residence time and only slightly dependent on the combustion gas temperature. Therefore, the thermal NO_x mechanism was thought to be largely suppressed and not a major contributor at the lower combustion temperatures. An alternate mechanism for NO_x formation is necessary to account for the observed behavior, including residence-time independence and very weak temperature dependence.

The NO_x concentrations for H_2/air mixtures plotted in Fig. 5 are found to be in agreement with those for CH_4/air mixtures. This implies that the NO_x in CH_4/air was formed by a mechanism without hydrocarbon reactions at $T_g < 1700$ K. As mentioned previously, P. C. Malte et al. proposed the nitrous oxide mechanism as an important mechanism of NO_x formation in lean-premixed combustion and suggested that the mechanism is a significant source, especially at temperatures below 1800 K [5,6]. They reported that their experimental data for NO_x emissions at combustion gas temperatures of 1650–1850 K in the jet-stirred combustion, shown in Fig. 4, could be accounted for by the modeling predictions, including the mechanism. Their predictions indicated that the dependence of NO_x formation on residence time becomes weak with decreasing temperature. This mechanism is expected to be an alternate mechanism for NO_x formation at $T_g < 1700$ K in the present impinging jet reactor.

Concluding Remarks

The mechanism of NO_x formation in lean-premixed, turbulent combustion of CH_4/air and H_2/air mixtures was investigated in an impinging jet reactor with minimum heat losses. The conclusions under the present operating conditions are as follows:

A change in the level of turbulence intensity did not result in a change in NO_x emissions.

At $T_g > 1700$ K, the NO_x formation was contributed primarily by the thermal mechanism. The NO_x emissions were proportional to the reactor residence time and sensitive to the combustion gas temperature. NO_x concentration for CH_4/air can be expressed as follows:

$$[NO_x] = \tau(5.96 \times 10^7) \exp(-31{,}500/T_g) \text{ (ppm)}$$

where τ is in ms and T_g in K.

At $T_g < 1700$ K, however, the NO_x concentrations were independent of residence time and only slightly sensitive to the combustion gas temperature. The NO_x concentrations for CH_4/air were in agreement with those for H_2/air. This implies that the NO_x was formed by a mechanism without hydrocarbon–nitrogen path. The nitrous oxide mechanism is expected to be an important mechanism for NO_x formation in the present impinging jet reactor.

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COMMENT

Dr. Gregory Smith, SRI International, USA. If the N_2O mechanism is responsible for the low temperature NO, one might expect more NO at leaner stoichiometry. Do you have information on how the NO varies with stoichiometry?

Author's Reply. Your comment and question are very interesting. In our experiments, the decrease in stoichiometric ratio of the premixture of CH_4/air or H_2/air resulted in a decrease in combustion gas temperature, leading to smaller NO_x emissions. We believe that the information on variation of NO_x emissions with stoichiometric ratio at a constant temperature would be more helpful to understanding of NO_x formation, particularly at leaner stoichiometric ratio.